

Abstract

University academic platforms function as static record-keeping systems, failing to proactively support students navigating complex degree requirements. **Orbis** is an intelligent academic support system bridging this gap using **Artificial Intelligence** and **Retrieval-Augmented Generation (RAG)**.

Core contributions:

- Organized **60,649 document chunks** into **125 high-quality semantic leaf nodes** via Shannon entropy classification and MiniBatchKMeans
- Extracted **178 actionable obligations** from 323 candidates via two-pass adversarial LLM review (**55% acceptance rate**)
- Generated **176 personalized assignments** for 4 student profiles in **59 minutes** of pipeline processing

Introduction

The Synchronization Gap:

Students miss critical deadlines not because information is unavailable, but because they do not know what to search for. Traditional Student Information Systems (SIS):

- Store data without interpreting it per student state
- Respond to queries but never alert proactively
- Apply uniform rules regardless of academic profile

Orbis shifts the academic paradigm from *static storage* to *autonomous orchestration* — an AI agent that evaluates each student's live academic state against institutional regulations and triggers timely, unprompted notifications.

Dataset

Knowledge base built from **80+ official IBU source URLs** scraping both English and Turkish mirrors of the university domain.

Dataset Property	Value
Total Document Chunks	60,649
Languages	Bilingual (EN + TR)
Regulatory Chunks Targeted	935
Final Taxonomy Leaf Nodes	125
Regulatory Leaf Nodes	29

The bilingual corpus requires careful deduplication. The Proactive Event System targets exclusively the **935 regulatory chunks** to prevent contamination from event pages and news feeds.

Theoretical Background

Shannon Entropy for URL Classification:

$$H = - \sum_{i=1}^n p_i \cdot \log_2(p_i)$$

High $H \Rightarrow$ semantically informative URL; Low $H \Rightarrow$ dynamic template (rejected).

Cosine Similarity for Semantic Retrieval:

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

L2-normalisation converts cosine to Euclidean distance, optimising KMeans centroid updates.

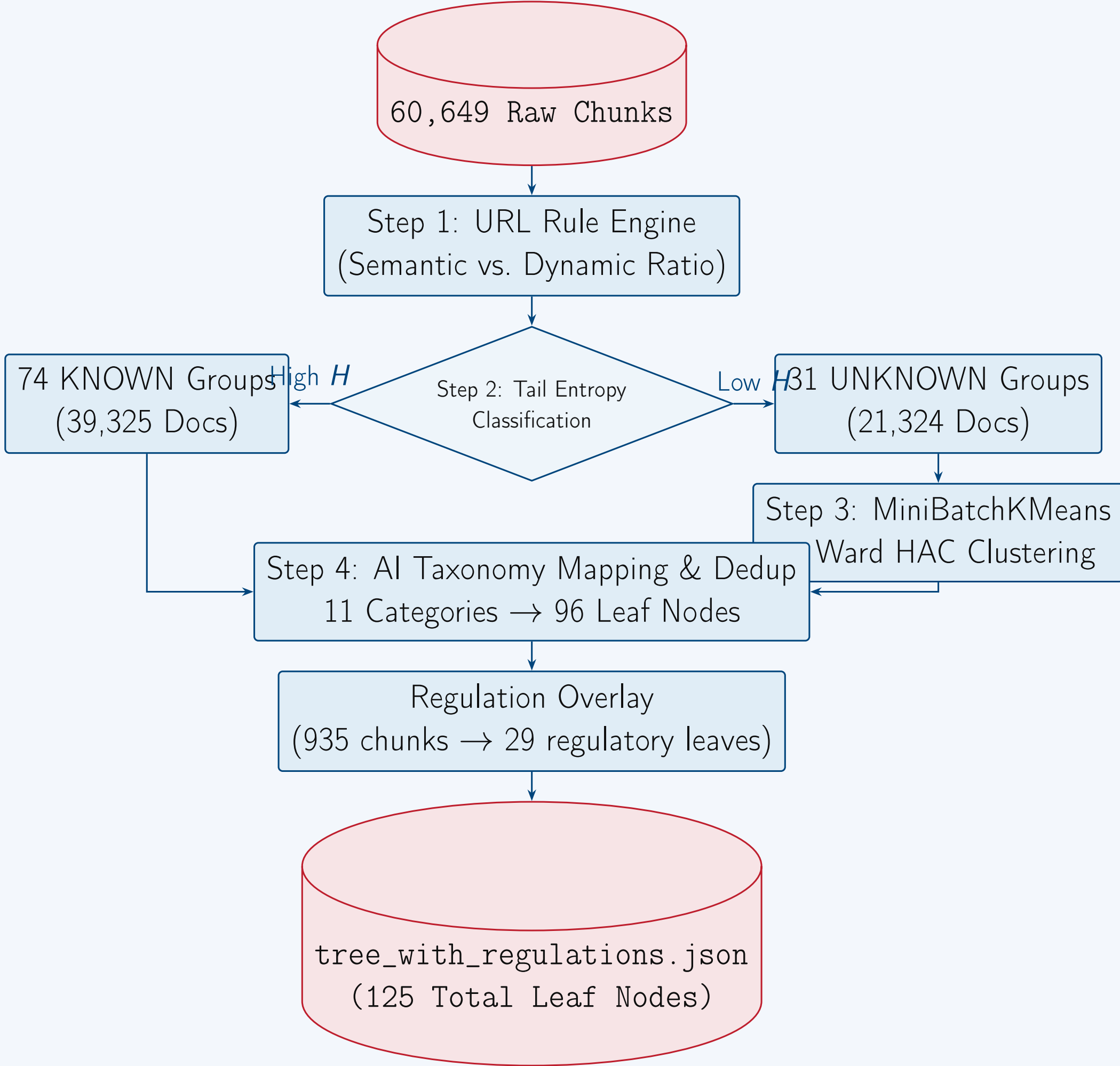
MiniBatchKMeans: $\mathcal{O}(n)$ linear clustering over 60k+ chunks, followed by Ward HAC on centroids to minimise within-cluster variance.

Extraction Acceptance Rate:

$$\text{Acceptance Rate} = \frac{178}{323} \approx 55\%$$

Proposed Methodology

Phase 1 — Data Categorisation & Taxonomy Pipeline:



Phase 2 — Orchestrator-Led Event Pipeline:

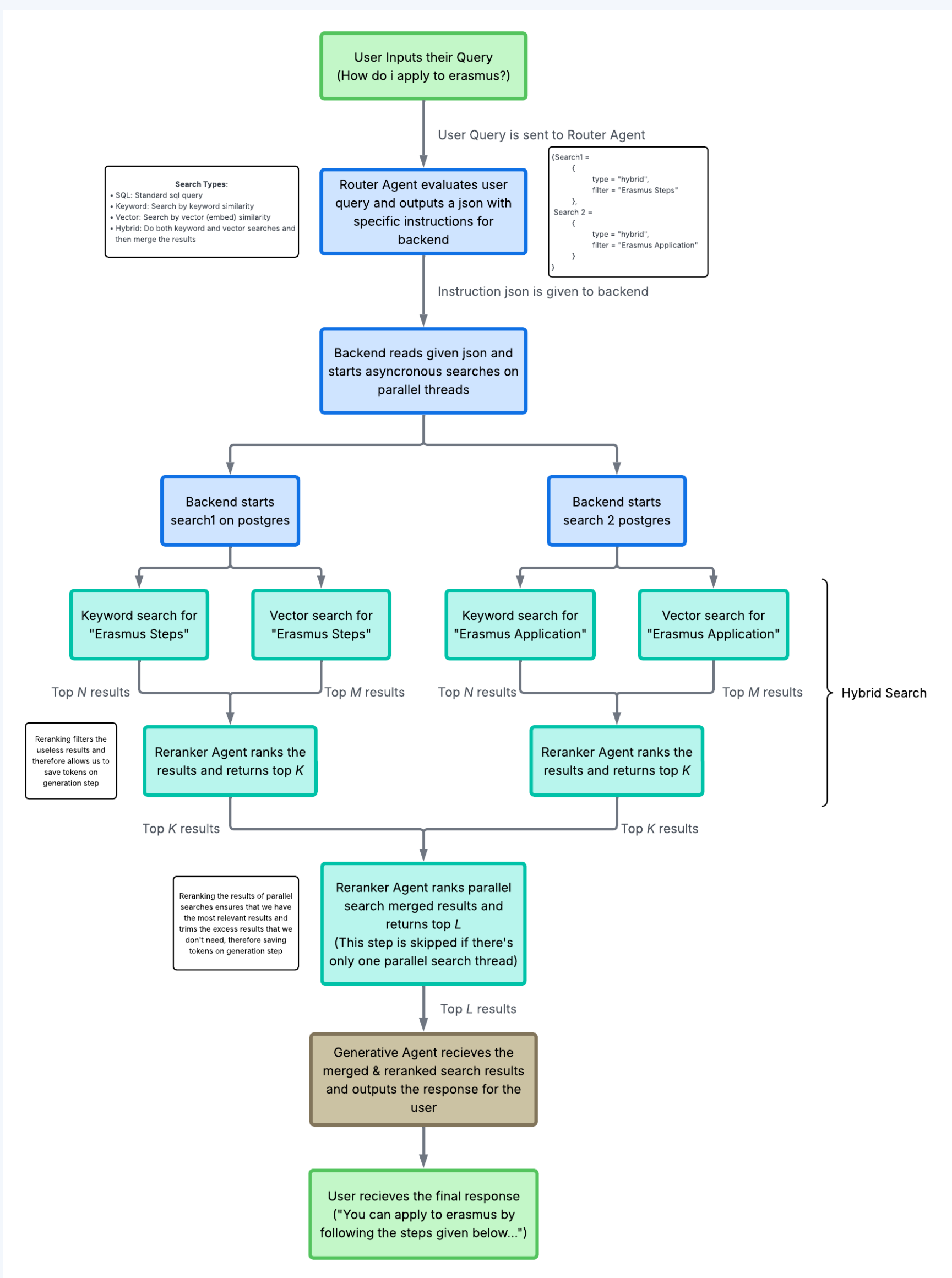
The centralised EventPipelineOrchestrator controls three state-less worker agents:

Agent	Responsibility
SearchAgent	Vector retrieval of regulatory chunks
ReasoningAgent	Per-student applicability evaluation
EventCreator	Normalised obligation construction

Two-Pass Adversarial LLM Review:

- Pass 1 — Extraction:** Full document evaluation; generates candidate rules with triggers, deadlines, and target student roles
- Pass 2 — Adversarial Fix:** Cross-references original document against Pass 1 output; removes stale items, strengthens weak wording, captures missed obligations

RAG System Architecture



Hybrid Assignment Matching — Two Parallel Channels:

- SQL Path (Deterministic):** Objective thresholds (GPA, credit count) evaluated instantly via Python logic — bypasses LLM entirely, minimising latency
- Contextual Path (LLM):** Complex rules evaluated against student extra_context JSON profile in parallel batches; LLM produces a human-readable justification per assignment

Double Reranking: Initial hybrid retrieval (keyword + vector) feeds two sequential cross-encoder reranking passes, achieving $MRR \approx 0.94$.

Experimental Setup

- Language:** Python 3.10+
 - API Framework:** FastAPI (microservices)
 - Async Processing:** Celery task queue
 - Database:** PostgreSQL + pgvector
 - LLMs:** OpenAI API + claude-opus-4-6
 - Frontend:** React (student dashboard)
 - Eval. basis:** Production DB logs
- Run ID: fadc244f...

Results: Extraction Quality Funnel

Pipeline Metric	Value
Source Documents Processed	80
KB Chunks Consumed	935
Candidate Rules Evaluated	323
Accepted Active Rules	178 (55%)
Rejected (Not Imperative)	145 (45%)
Total Pipeline Runtime	59 min

Of 145 rejected candidates, **128** were flagged for “*not imperative enough*” wording — a strictly intended outcome of the Specificity Filter, guaranteeing students receive only actionable obligations.

Results: Assignment Distribution

176 total assignments for 4 student profiles:

Student	Tot.	High	SQL	CTX
Mehmet (Yr4)	60	38	2	58
Ali (Yr2, Prob.)	42	30	8	34
Zeynep (Yr3, Int.)	38	29	2	36
Ayşe (Yr3, DM.)	36	25	3	33
Total	176	122	15	161

Contextual example — Zeynep (*internship required, no company found yet*): System generated HIGH urgency task: **“Submit the Internship Contract Cover Form at least 10 working days before your start date”** — reasoning that the obligation activates upon company placement.

SQL example — Ali (*GPA 1.60*): Deterministic SQL path instantly matched 8 rules against $gpa < 2.00$ — zero LLM calls needed.

RAG Performance Metrics

RAG Metric	Score
Routing Accuracy	$\approx 92\%$
Context Recall (Retrieval)	$\approx 95\%$
Mean Reciprocal Rank (MRR)	≈ 0.94
Semantic Accuracy	$\approx 98\%$
Answer Token F1-Score	$\approx 78\%$

Smart Expansion retrieves full context beyond 150-word chunk limits for procedural queries. **CSV Compacting** prevents 30k token overflow for bulk SQL lookups — both contributing to 98% semantic accuracy.

Conclusion & Future Work

Demonstrated:

- ✓ 60,649 chunks $\rightarrow$ 125 semantic leaves (Entropy + KMeans)
- ✓ 178 actionable rules with 55% Specificity Filter
- ✓ 176 personalized assignments in <1 hour
- ✓ 98% semantic accuracy across the RAG pipeline

Future Work:

- Direct SIS API integration for automated student context ingestion
- Rejection learning to continuously refine LLM extraction models
- Multi-university scalability testing